

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Modular Unit 1

Adaptive Classified

OL-T54 · Histograms

Questions

Student book · questions with writing space

■ 19 pages

FAMILY MAP

Histograms

F01	Interpret histogram areas	11 Q 37 marks
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F02	Construct and complete histograms	5 Q 13 marks
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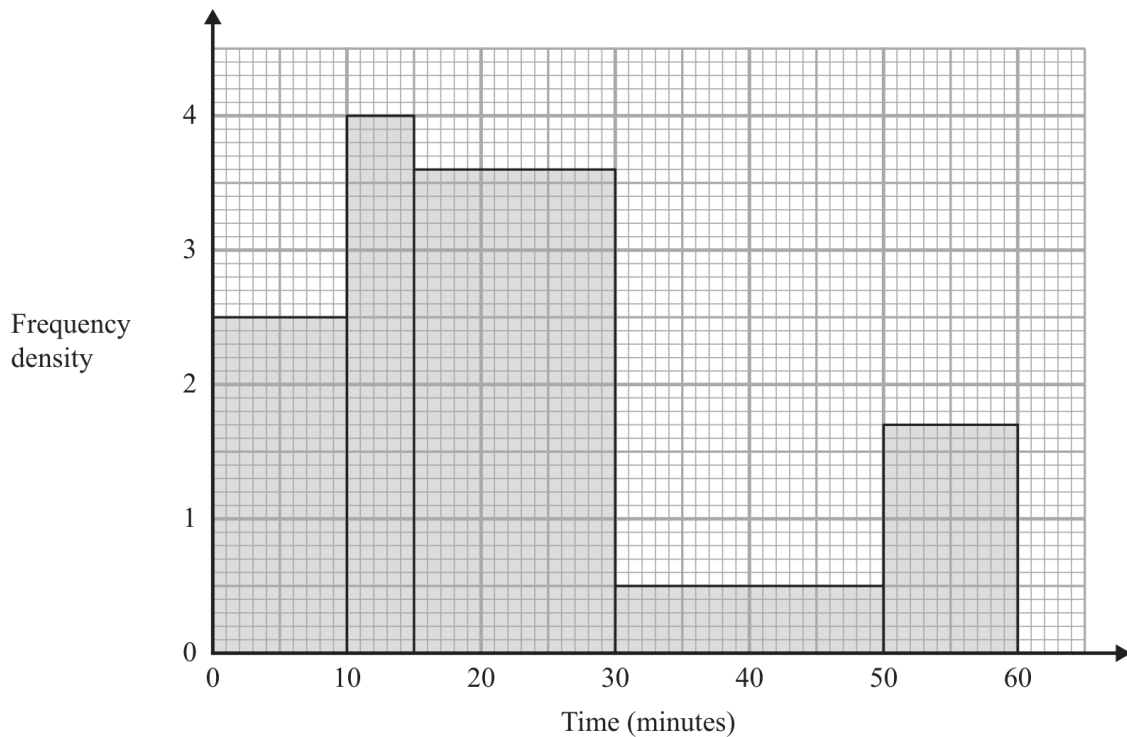
F03	Use the frequency-density model	1 Q 3 marks
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Working Unit 12

4MA1-1H Q17

UNIT 1
3 marks

- 17 The histogram shows information about the times, in minutes, that some people were in a shop.



Work out an estimate for the proportion of these people who were in the shop for more than 40 minutes.

(Total for Question 17 is 3 marks)

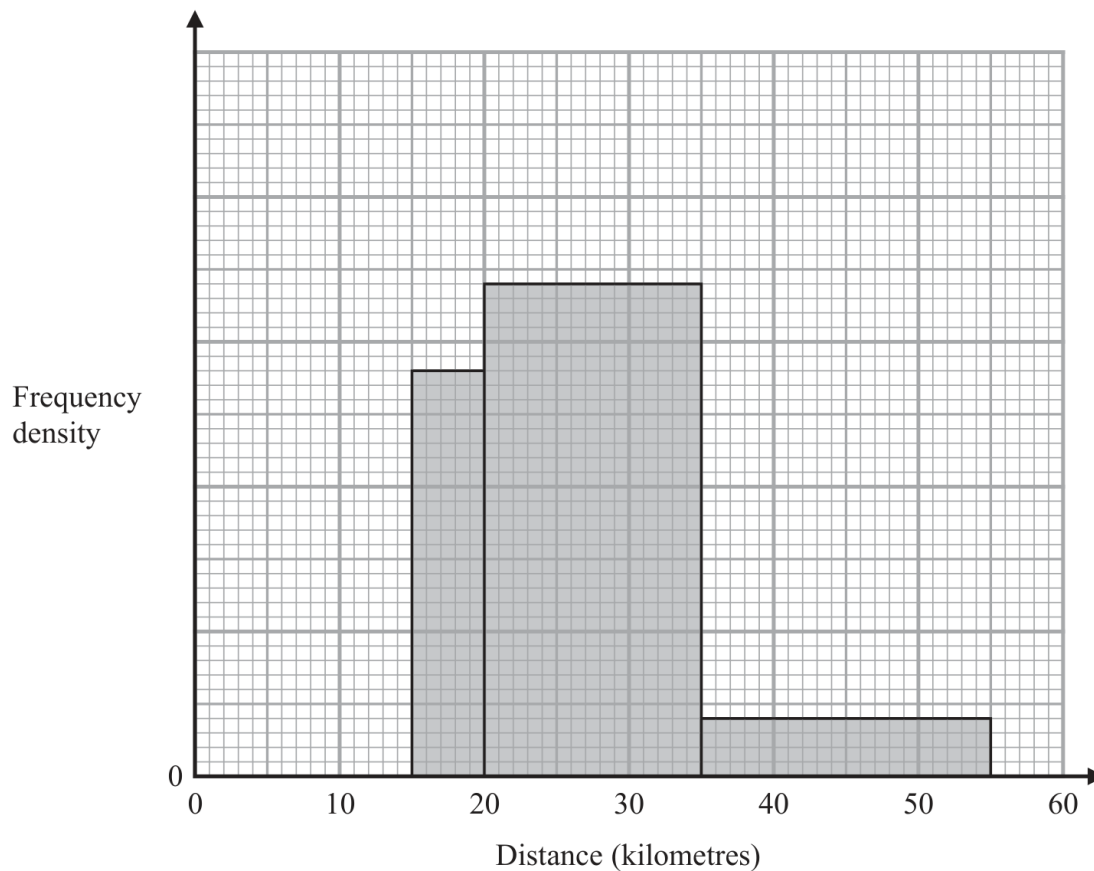
Question 15

4MA1-1H Q22

MODULAR UNIT 1

3 marks

- 22 The incomplete histogram shows some information about the distances, in kilometres, that 100 adults ran last week.



All of the adults ran at least 5 kilometres.
 None of the adults ran more than 55 kilometres.
 14 adults ran between 15 kilometres and 20 kilometres.

Complete the histogram.

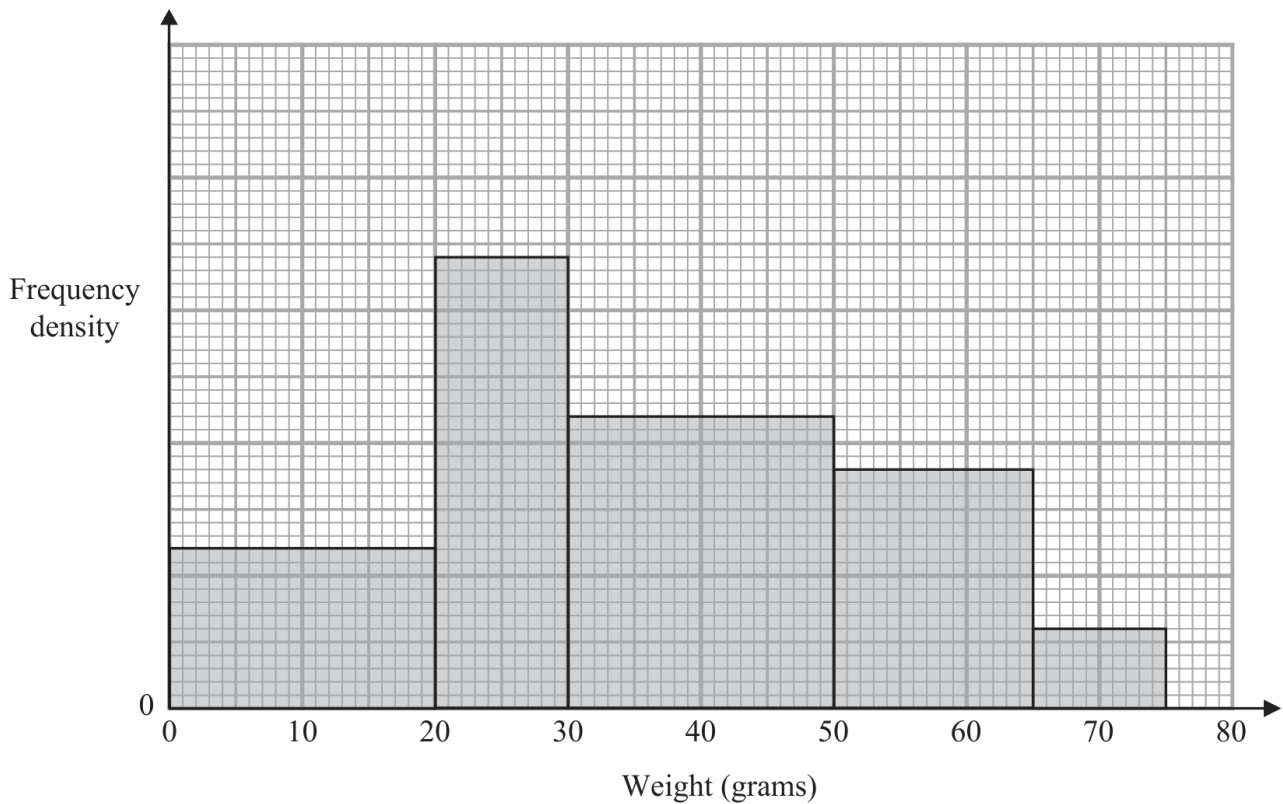
Question 13

4MA1-1H Q17 demand

MODULAR UNIT 1

3 marks

17 The histogram gives information about the weights, in grams, of some oranges in a box.



24 of these oranges weigh less than 20 grams.

Medium oranges weigh between 35 grams and 55 grams.

Work out an estimate for the number of medium oranges in the box.

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Question 10

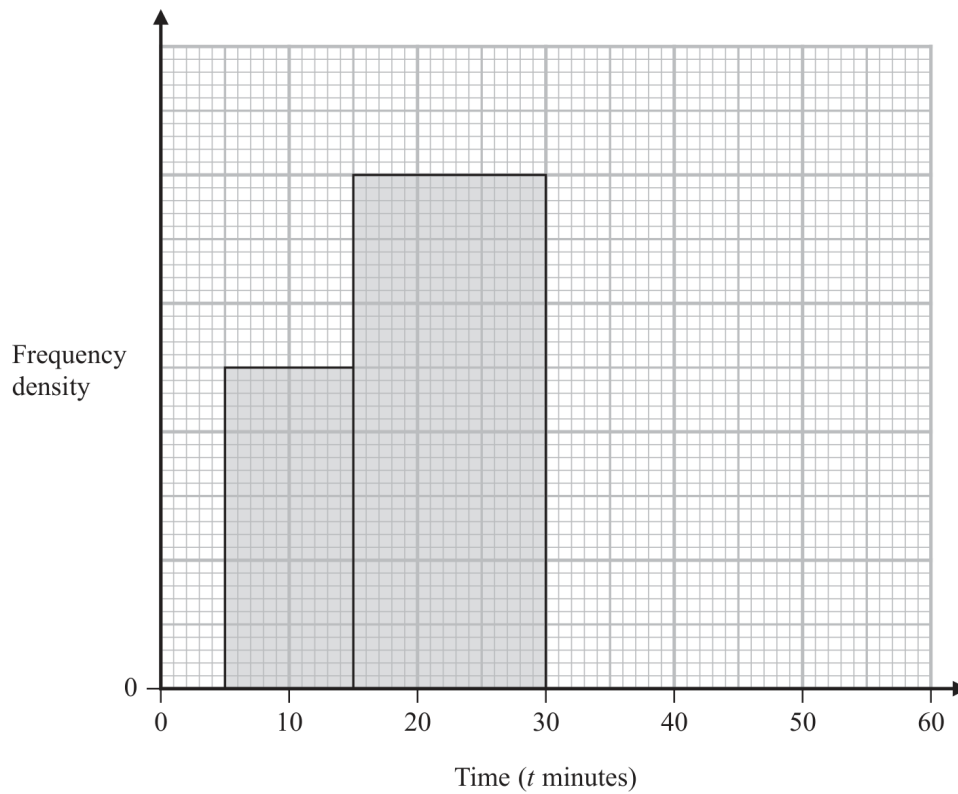
4MA1-1H Q18

MODULAR UNIT 1

4 marks

- 18 The incomplete table and incomplete histogram give information about the times, in minutes, that 140 people waited at a station for a train.

Time (t minutes)	Frequency
$0 < t \leq 5$	23
$5 < t \leq 15$	
$15 < t \leq 30$	
$30 < t \leq 40$	18
$40 < t \leq 60$	14



Complete the table and the histogram.

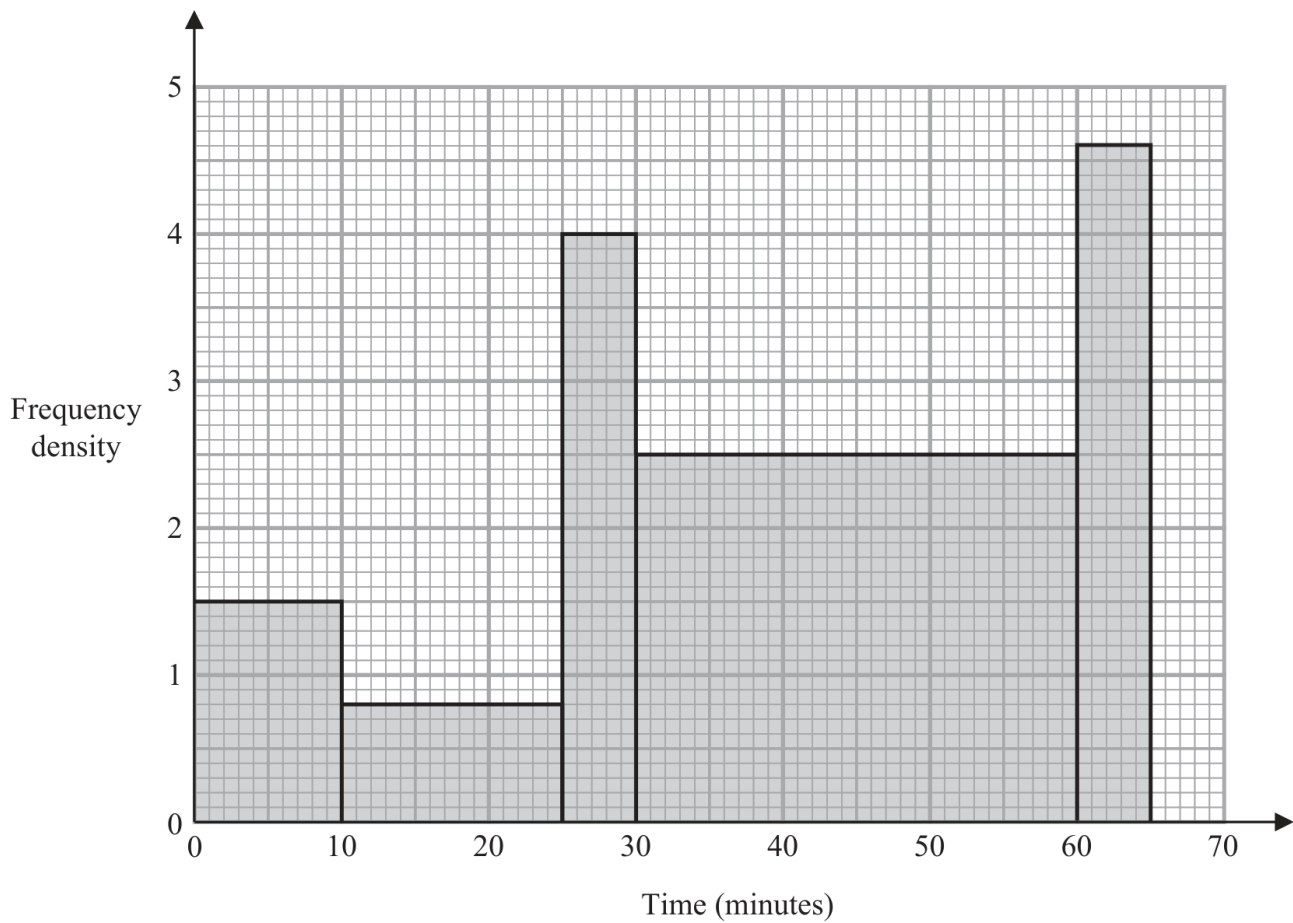
Question 15

4MA1-1H Q18 M01

MODULAR UNIT 1

4 marks

- 18 The histogram shows information about the times some students took to complete a puzzle.



Work out an estimate for the fraction of these students who took between 20 minutes and 60 minutes to complete the puzzle.

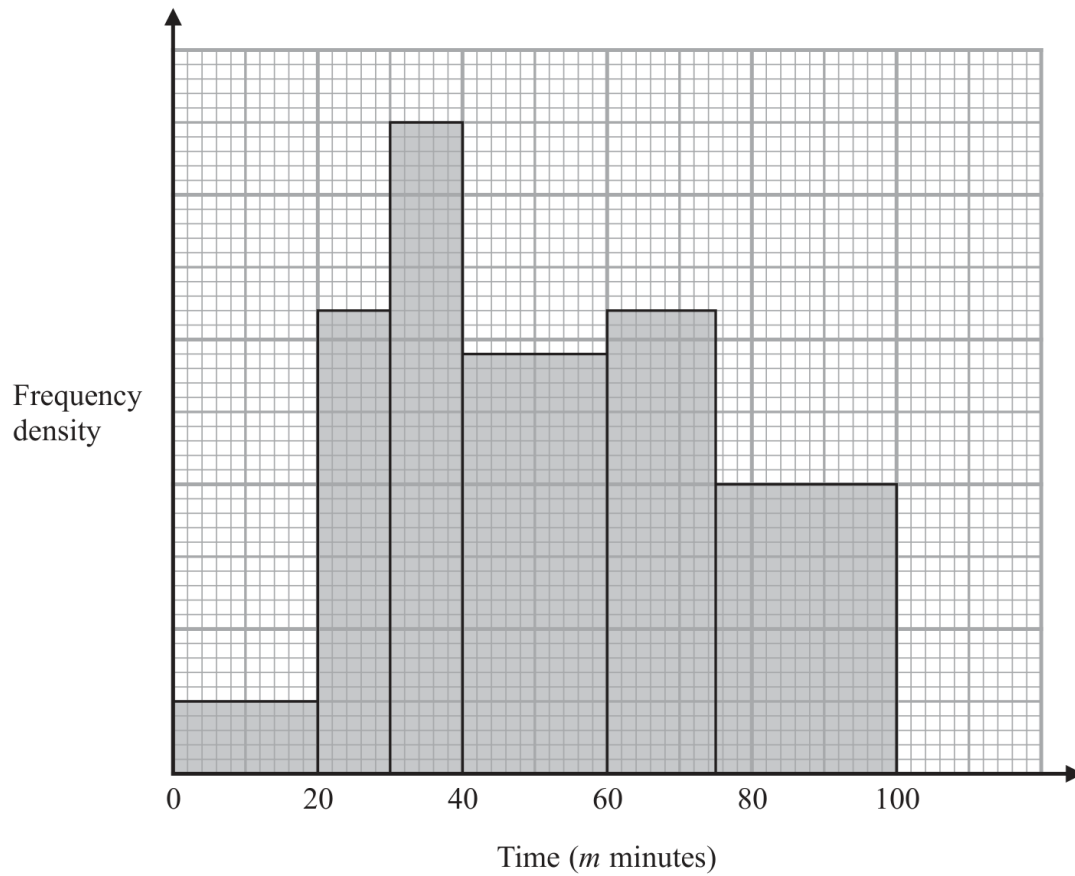
Question 14

4MA1-1H Q21 Q21

MODULAR UNIT 1

3 marks

- 21 The histogram shows information about the total time, m minutes, taken by each child in a school to walk to school every day for one week.



There are no children for whom $m > 100$

There are 10 children for whom $m \leq 20$

Work out an estimate for the number of children for whom $50 < m \leq 80$

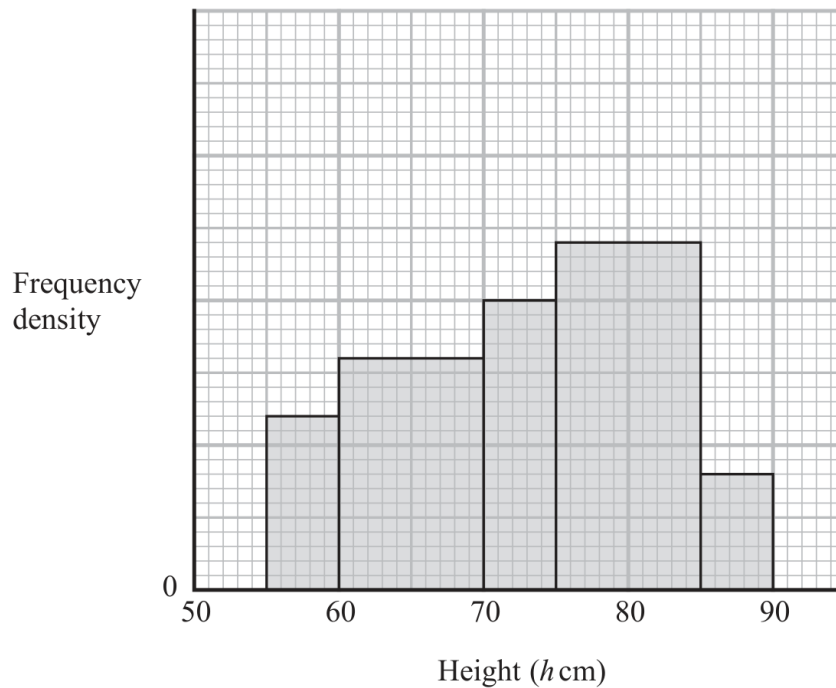
Question 26

4MA1-2H Q20

MODULAR UNIT 1

4 marks

20 The histogram gives information about the heights, h cm, of some tomato plants.

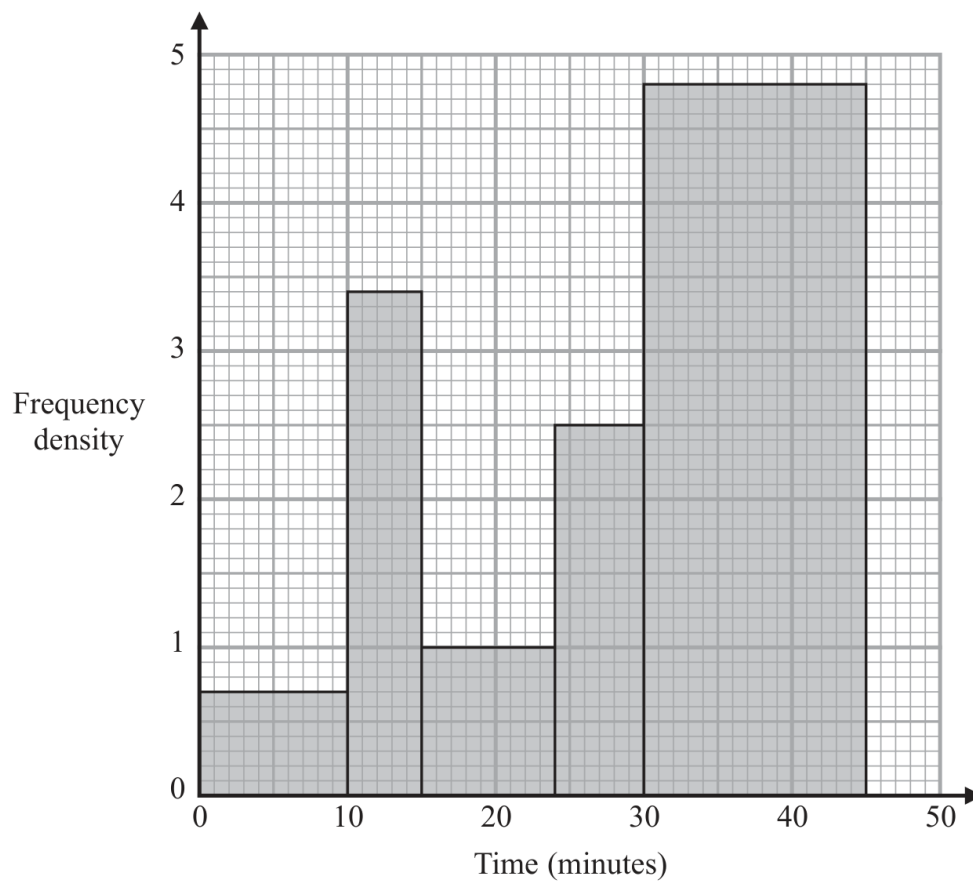


There are 12 tomato plants for which $75 < h \leq 85$

One of the tomato plants is selected at random.

Find an estimate for the probability that this tomato plant has a height greater than 82.5 cm

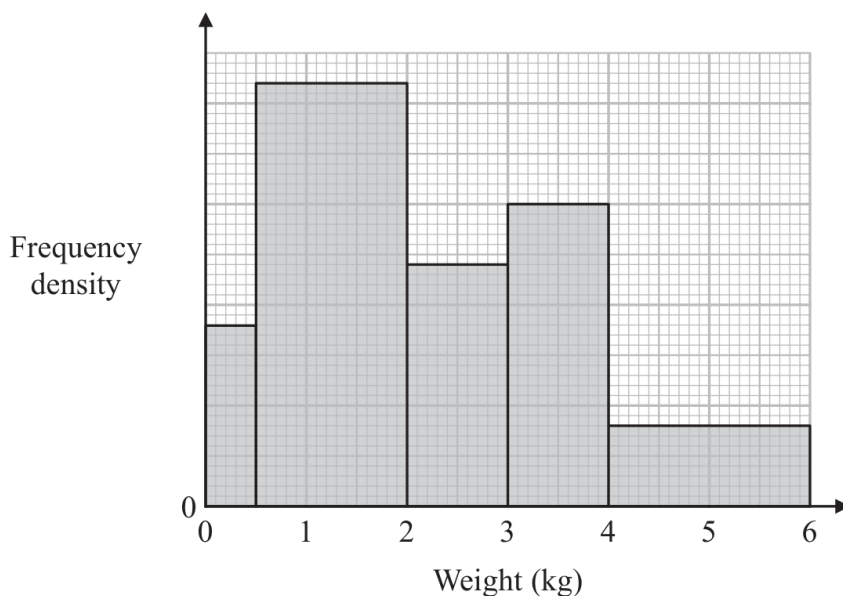
18 The histogram gives information about the times, in minutes, that some customers spent in a supermarket.



- (a) Work out an estimate for the proportion of these customers who spent between 17 minutes and 35 minutes in the supermarket.

15 A postman records the weight of each parcel that he delivers.

The histogram shows information about the weights of all the parcels that the postman delivered last Monday. No parcels weighed more than 6 kg.



63 of the parcels that the postman delivered last Monday each had a weight between 0.5 kg and 2 kg.

(a) Work out the total number of parcels the postman delivered last Monday.

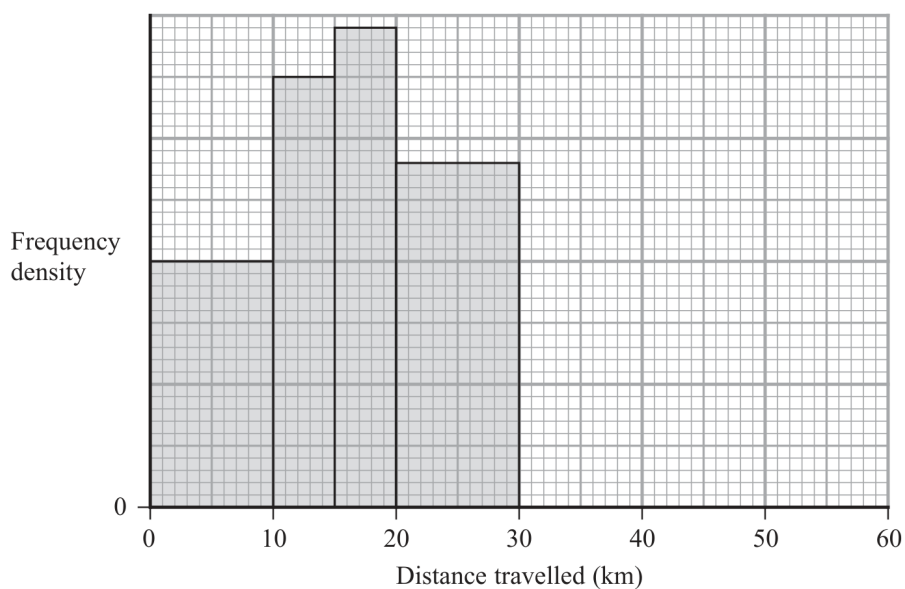
(3)

The postman picks at random two of the records of the parcels he delivered last Monday.

(b) Work out an estimate for the probability that each parcel weighed more than 2.25 kg.

- 18 The table and histogram give information about the distance travelled, in order to get to work, by each person working in a large store.

Distance (d km)	Frequency
$0 \leq d < 10$	40
$10 \leq d < 15$	
$15 \leq d < 20$	
$20 \leq d < 30$	
$30 \leq d < 60$	30



Using the information in the table and in the histogram,

- (a) complete the table,

(2)

- (b) complete the histogram.

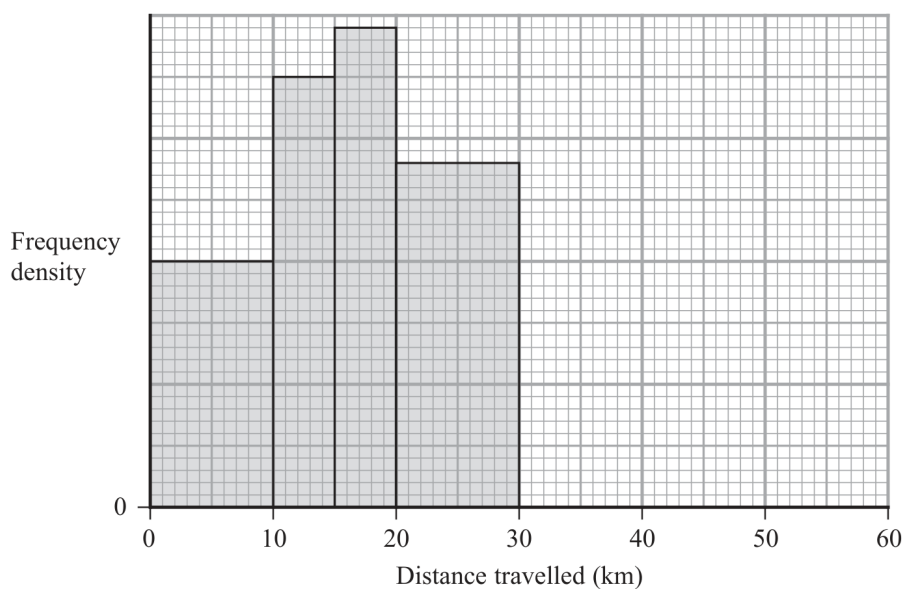
(1)

One of the people working in the store is chosen at random.

- (c) Work out an estimate for the probability that the distance travelled by this person, in order to get to work, was greater than 25 km.

- 18 The table and histogram give information about the distance travelled, in order to get to work, by each person working in a large store.

Distance (d km)	Frequency
$0 \leq d < 10$	40
$10 \leq d < 15$	
$15 \leq d < 20$	
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$30 \leq d < 60$	30



Using the information in the table and in the histogram,

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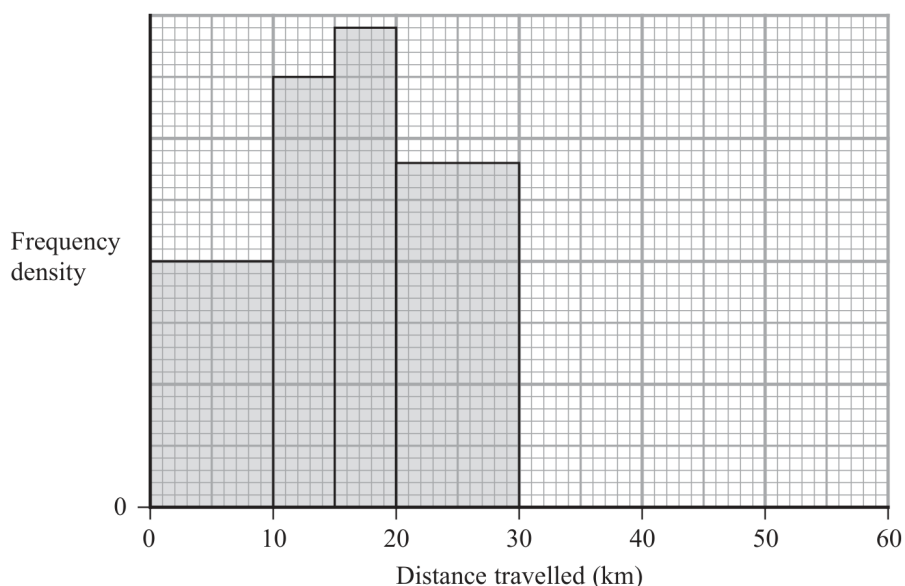
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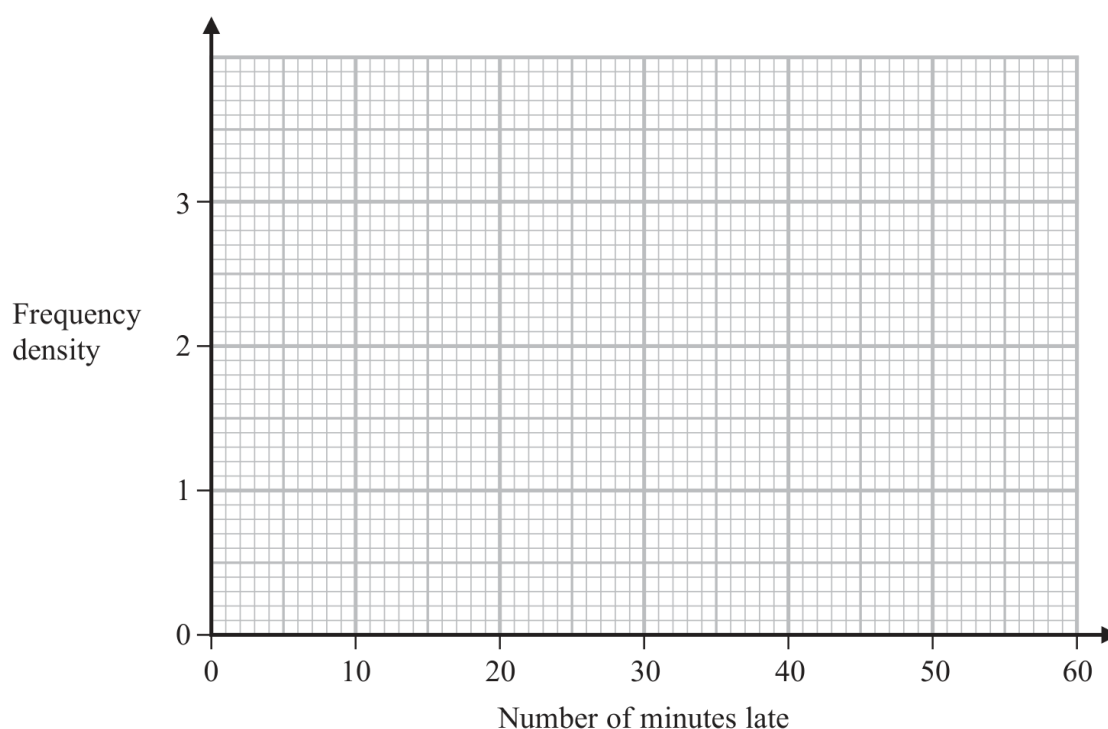
- (c) Work out an estimate for the probability that the distance travelled by this person, in order to get to work, was greater than 25 km.

14 Manuel collected information about the flights that arrived late at an airport last month.

The table gives information about the number of minutes that these flights were late.

Minutes late (L minutes)	Frequency
$0 < L \leq 10$	8
$10 < L \leq 15$	13
$15 < L \leq 25$	19
$25 < L \leq 40$	24
$40 < L \leq 60$	6

(a) On the grid, draw a histogram for this information.



(3)

Manuel selected at random a flight that was late by 25 minutes or less from his results.

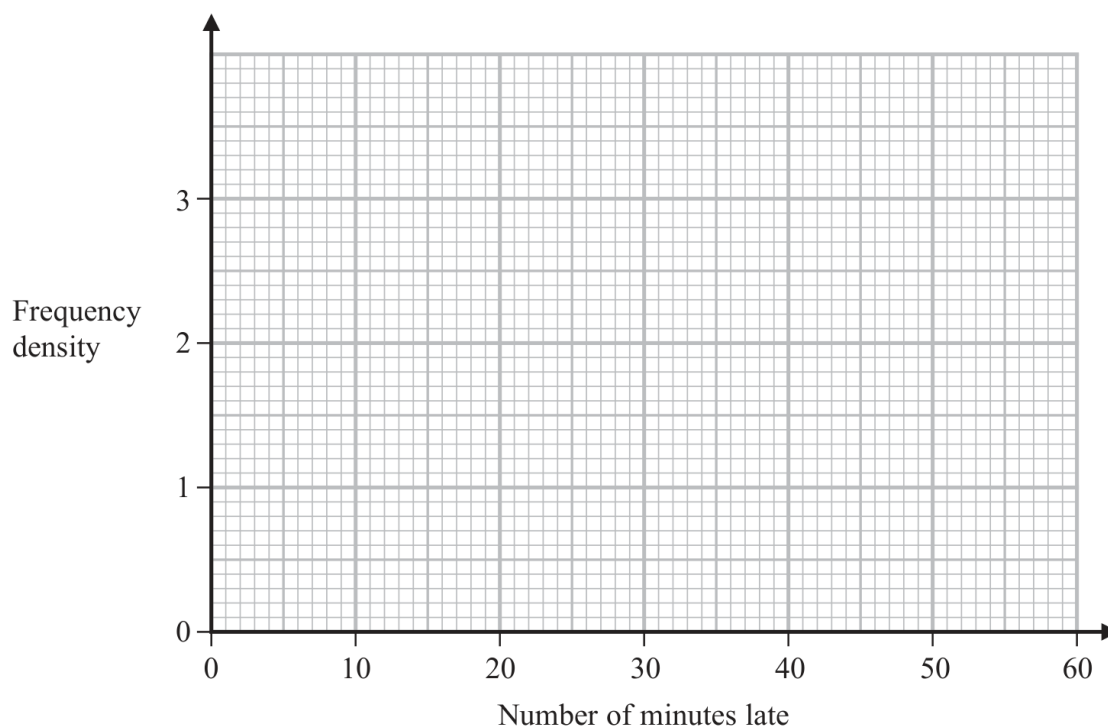
(b) Work out an estimate for the probability that this flight was late by 5 minutes or less.

14 Manuel collected information about the flights that arrived late at an airport last month.

The table gives information about the number of minutes that these flights were late.

Minutes late (L minutes)	Frequency
$0 < L \leq 10$	8
$10 < L \leq 15$	13
$15 < L \leq 25$	19
$25 < L \leq 40$	24
$40 < L \leq 60$	6

(a) On the grid, draw a histogram for this information.



(3)

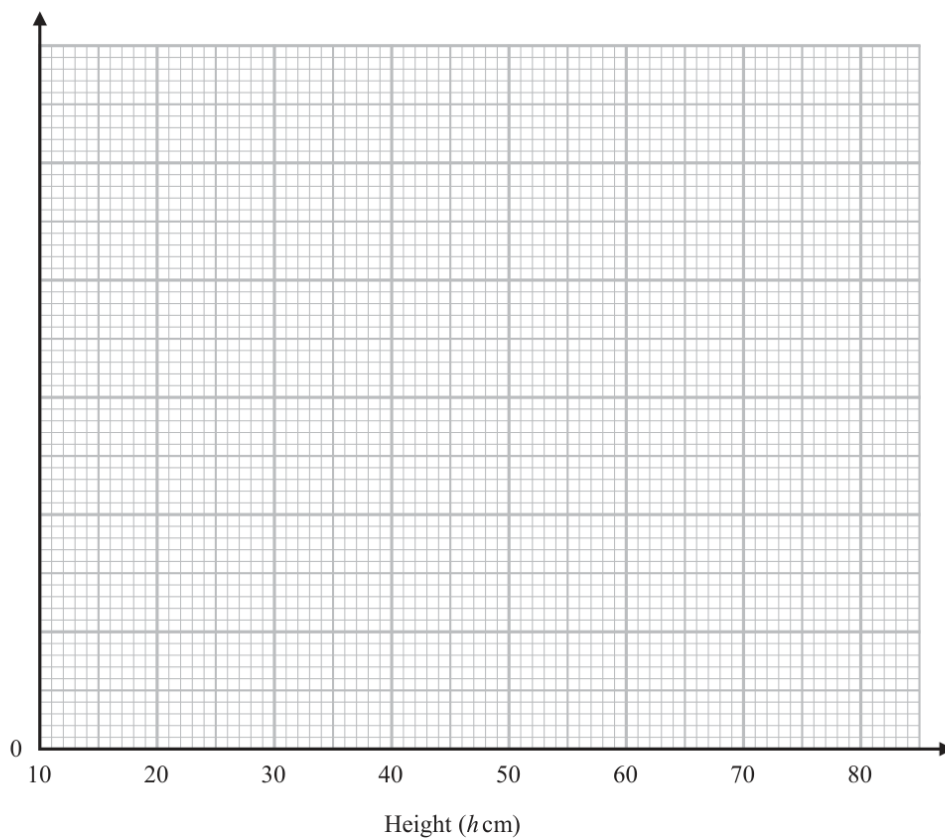
Manuel selected at random a flight that was late by 25 minutes or less from his results.

(b) Work out an estimate for the probability that this flight was late by 5 minutes or less.

18 The table gives information about the heights, in centimetres, of some plants.

Height (h cm)	Frequency
$10 < h \leq 20$	35
$20 < h \leq 35$	45
$35 < h \leq 50$	75
$50 < h \leq 70$	40
$70 < h \leq 80$	8

(a) On the grid, draw a histogram for this information.



(b) Work out an estimate for the number of these plants with a height greater than 40 cm.

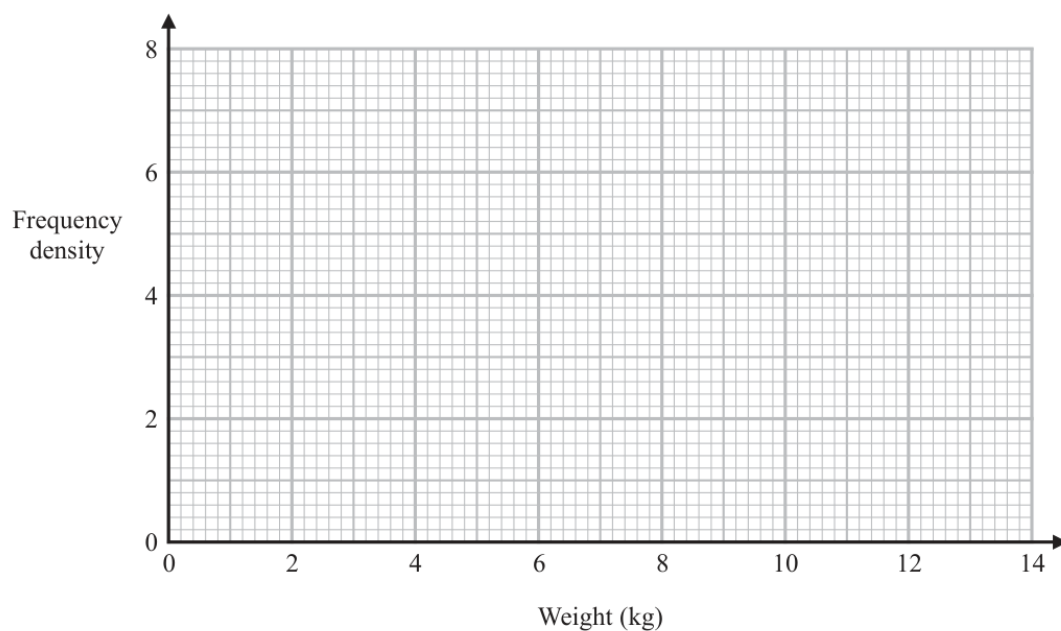
.....
(2)

(Total for Question 18 is 5 marks)

- 18 The table gives information about the weights, in kg, of the parcels that Pedro delivers on Monday.

Weight (w kg)	Frequency
$0 < w \leq 2$	12
$2 < w \leq 3$	7
$3 < w \leq 6$	15
$6 < w \leq 9$	12
$9 < w \leq 14$	9

- (a) On the grid, draw a histogram for this information.



(3)

One of the parcels that Pedro delivered on Monday is chosen at random.

- (b) Using the information in the table, find an estimate for the probability that this parcel weighs more than 7 kg.

A histogram showing the frequency density of time taken to complete a 1000m race. The x-axis is labeled 'Time (minutes)' and ranges from 0 to 60 with major grid lines every 10 units and minor grid lines every 2 units. The y-axis is labeled 'Frequency density' and ranges from 0 to 10 with major grid lines every 2 units and minor grid lines every 0.5 units. The histogram consists of five bars with the following widths and heights:

Time Interval (minutes)	Frequency Density
0 - 10	4.0
10 - 15	8.0
15 - 30	6.0
30 - 40	2.0
40 - 60	1.0

Work out an estimate for the number of these trains that arrived at least 25 minutes late.